CORMAC's Comments for Health Technology Ecosystem (code CMS-0042-NC)

Centers for Medicare & Medicaid Services (CMS)

Submitted To:	Offeror: CORMAC Corporation	Point of Contact & Official Authorized to Bind CORMAC
Agency: Centers for Medicare & Medicaid Services (CMS), **Address**: CMS-0042-NC, P.O. Box 8013, Baltimore, MD 21244-8013. **Contact**: HealthTechRFI@cms.hhs.gov	cormac HEALTH DATA ANALYTICS CMMI ML3 | ISO 9001 | ISO 20000 | SB | WOSB 161 Fort Evans Road, NE Suite 250 Leesburg, VA 20176 MAIN: (443) 864 5880 FAX: (443) 319 5483 http://www.cormac-corp.com	Dominic Raj Phone: (703) 568-1292 Dominic.Raj@cormac-corp.com CREPES AI

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1 Executive Summary

Since 2002, **CORMAC Corporation** has delivered mission-aligned technology solutions across the federal landscape with a focus on data-driven transformation, digital enablement, and AI innovation. In direct alignment with the CMS and ASTP/ONC objectives outlined in this Request for Information (RFI), CORMAC brings deep experience in supporting the public sector with advanced tools that expand patient access to digital health capabilities, reduce administrative burden, and accelerate interoperability across healthcare systems.

Our AI practice, **CREPES.AI**, plays a pivotal role in developing scalable digital health infrastructure. This platform includes multiple SaaS products designed to streamline critical government workflows through artificial intelligence, natural language processing (NLP), and data visualization. CREPES.AI is hosted on FedRAMP-compliant AWS infrastructure and is engineered to automate labor-intensive processes and enable decision-making that improves outcomes and operational efficiency.

We are actively collaborating with the **DHS Procurement Innovation Lab (PIL)** on two of our SaaS products:

- **CREPES.AI Past Performance (PP)** for automated vendor performance assessments, and
- **CREPES.AI Market Research (MR)**, which has been approved for use across **all federal agencies** to support acquisition planning through automated sourcing and market analysis.

Our modular SaaS offerings are well-positioned to support CMS in achieving its vision of a **patient-centric learning health system** by enabling more equitable, timely, and intelligent access to digital tools for Medicare beneficiaries and their caregivers. CORMAC offers flexible provisioning models—including standalone SaaS or integrated service-based deployments—to meet agencies where they are on their modernization journey.

Our company's core capabilities directly support CMS's goals of improving data availability, expanding interoperability, encouraging digital health innovation, and reducing provider and patient burden. As CMS continues to lead efforts that foster trusted digital health ecosystems and value-based care transformation, CORMAC stands ready to contribute solutions that meet today's challenges while supporting future policy frameworks and infrastructure needs.

2 Technology Vendors, Data Providers, and Networks

2.1 Ecosystem

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

To foster innovation that improves the health and wellbeing of Medicare beneficiaries and their caregivers, CMS must intentionally cultivate an ecosystem that attracts and supports developers. CORMAC offers the following short-term and long-term recommendations that focus on expanding data access, enhancing the developer experience, and incentivizing engagement through strategic programs and partnerships. These steps will help CMS stimulate broad-based interest in building impactful digital health tools.

Short-Term Actions (Next 2 Years)

1. Expand Access to Anonymized Data for Innovation:

CMS should establish a centralized, anonymized Data Lake to supplement the existing Blue Button API. This environment will provide developers with broad, structured datasets in a standardized and developer-friendly format. Doing so will reduce technical barriers, lower integration costs,

and support the development of AI-powered solutions tailored to Medicare beneficiaries and caregivers.

2. Launch Developer-Centric Innovation Events:

CMS is encouraged to host innovation-oriented events such as Hackathons, Developer Challenges, and design competitions. These activities will spark interest, promote creative problem-solving, and engage a wide range of technical talent. CMS could also distribute sample datasets and problem statements at conferences or through platforms like Amazon Mechanical Turk to further widen participation.

3. Incentivize Through Gamification and Recognition:

To foster sustained engagement, CMS should implement gamification features that include certifications, digital badges, and public recognition for developer contributions. CMS is also encouraged to promote team-based participation by pairing senior and junior developers, thereby facilitating mentorship and skill development within the community.

4. Build a Robust Developer Experience Layer:

CMS should enhance the usability of its development environment by building a comprehensive Developer Experience Layer. This may include curated API bundles organized by use case (e.g., care navigation, medication management), sample FHIR resources, and pre-built consent flows. These enhancements will simplify integration, reduce ramp-up time, and increase the appeal of building on CMS platforms.

5. Provide Data Access Harnesses and Developer Sandboxes:

CMS should offer robust sandbox environments and test harnesses with high-quality synthetic data that mimic real-world CMS systems. These tools would allow developers to simulate interactions such as claims processing, provider transitions, and drug formulary changes. By providing realistic testing conditions, CMS will help developers build and refine prototypes more effectively.

6. Data Access Test Harnesses and Developer Sandboxes:

Providing synthetic data environments that simulate real-world CMS responses, including claim lifecycle events, provider transitions, drug formularies, would reduce developer ramp-up time and increase prototype realism.

Long-Term Actions (Beyond 2 Years)

Over the long term, CMS should form partnerships with other federal agencies (e.g., VA, NIH), state health systems, and private sector organizations to expand the scope and richness of accessible datasets. These collaborations will enable the development of more comprehensive and personalized digital tools that support improved care coordination, equity, and health outcomes for Medicare beneficiaries and their caregivers.

By advancing these strategic actions, CMS builds a future-ready developer ecosystem that fuels meaningful innovation. CORMAC stands ready to support this vision and work collaboratively to foster solutions that elevate the health, wellbeing, and care experiences of America's Medicare population.

TD-2. Regarding CMS Data, to stimulate developer interest—

a. What additional data would be most valuable if made available through CMS APIs?

In order to provide a complete health profile of the beneficiary, In addition to Part A, B and D claims data, CMS could provide access to the following data via APIs

- Part C - Medicare Advantage claims data
- ACA claims data
- Prior Authorization and Utilization Management Data – useful for Apps that provide information related to complex procedures with possible delays
- Any historical data related to Out-of-Pocket Cost Estimates
- Include Provide quality data such as Affiliations, Star ratings

b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

CMS could establish a partnership with other Health agencies such as FDA, NIH, AHRQ, VA; State Medicaid agencies and private health practices to expand the data sources. Some examples are given below:

1. AHRQ Patient safety and survey data for better insights into patient safety aspects
2. FDA – Drug adverse events for better insights on drug interactions
3. VA – claims data related to Veterans
4. CDC – data such as National Healthcare Safety Network (NHSN) for healthcare-associated infections and prevention efforts
5. NIH -some examples are age-related diseases and the aging process from National Institute on Aging (NIA) resources; Clinical Trials Databases: on new treatments and interventions for various diseases. Medicare beneficiaries might find information on clinical trials relevant to their specific health conditions
6. Social Determinants of Health (SDOH) Data available outside CMS such as , linked public datasets (e.g.: Census data)
7. Clinical Data from EHRs via FHIR APIs such as vitals, lab results

c. What obstacles prevent accessing these data sources today?

Accessing the data is challenging because each agency's IT system safeguards its own data in isolation. We recommend breaking down these data silos to enable secure, privacy-preserving access for the developer and research communities, without compromising any privacy or security regulations.

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

1. Smart on FHIR APIs - SMART on FHIR, provides a health app interface based on open standards including HL7's FHIR, OAuth2, and OpenID Connect. SMART on FHIR applies a set of "profiles" that provide developers with expectations about the vocabularies that will be used to express medications, problems, labs, and other clinical data.
    a. Key features include:
        i. Developers work directly with concrete data types such as allergy, prescription, or condition, not abstract types such as entity, actor, or role
        ii. SMART specifications are 'opinionated,' making up-front choices about how data are represented so that developers know what to expect. For instance, every problem in a SMART record is associated with a SNOMED CT disorder code
        iii. SMART defines a limited set of broadly applicable data types, rather than permitting a proliferation of interface-specific definitions

2. Provider Directory API via FHIR – up-to date information on the providers such as exclusions and terminations, network participation, location, associated labs and specialties
3. APIs related to benefits and coverage, prior authorization requirements
4. APIs on provider quality measures provider- or facility-level performance and outcomes data
5. Clinical Data APIs (Bulk FHIR or TEFCA-based)

2.2 Digital Identity

TD-3. Regarding digital identity implementation:

a. What are the challenges and benefits?

Challenges include inconsistent adoption of credentialing protocols across federal and commercial systems, leading to fragmented patient experiences and limited reuse of digital identity credentials. Many health IT systems still default to proprietary login models, requiring patients and caregivers to maintain multiple credentials for different providers or portals. Benefits of standardized digital identity credentials include improved trust between systems, streamlined patient access, and reduced overhead for identity lifecycle management, particularly in use cases involving multi-institutional data sharing, such as care coordination, TEFCA participation, or Blue Button API authorization.

b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?

Mandating identity credentials would strengthen end-user authentication without requiring each provider or vendor to individually manage risk evaluation frameworks. This would reduce credential sprawl, improve auditability, and standardize access control enforcement across exchange ecosystems, especially those leveraging SMART on FHIR or bulk FHIR API endpoints. Additionally, stronger credentialing would improve assurance during sensitive operations such as claims-based access or patient-generated data submission.

c. What impact would mandatory use of the OpenID Connect identity protocol have?

OpenID Connect (OIDC) adoption would unify the identity abstraction layer across disparate applications, particularly for FHIR-based API ecosystems. It would simplify authorization code flow across vendors and enable broader reuse of authenticated identity assertions for delegated app access, including enabling patients to consent to third-party access via a credentialed workflow.

2.3 Technical Standards and Certification

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

1. CMS could investigate and/or implement any of the following options: Regulatory Levers, Financial and Programmatic Incentives, Transparency and Accountability, Technical and Infrastructure Support, Vendor and Provider Engagement, and Discouraging Proprietary Lock-In.
2. Overall, Blue Button 2.0 is a specific CMS initiative focused on Medicare claims, whereas open, standards-based APIs, such as SMART on FHIR is a foundational interoperability standard designed for widespread clinical data exchange across the healthcare ecosystem.
3. CMS could partner with other agencies that have built APIs with public SDKs such as ONC/HL7 to announce a cooperative initiative that would have benefits for both programs.

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

A nationwide provider directory of FHIR endpoints would significantly enhance health information exchange and interoperability across the U.S. healthcare system. Also, this will minimize the Fraud related to provide identity. The following table summarize the benefits:

Stakeholder	Benefits
Patients	Better data access & portability: Patients can more easily retrieve their health records from various providers through apps that locate the correct FHIR endpoints. Reduced data silos: Seamless record transfer enables longitudinal health records across health systems.
Providers	Streamlined care coordination: Providers can quickly locate and query FHIR endpoints for outside patient records. Less administrative burden: Less manual processes like fax or phone for records exchange. Supports TEFCA & 21st Century Cures compliance: Helps meet regulatory requirements for information blocking and interoperability.
Payers	Efficient data retrieval: Automates access to claims, clinical data, and prior authorization workflows. Better risk adjustment & analytics: Real-time data exchange supports value-based care and population health. Improved provider network management: Helps verify participating organizations' interoperability readiness.

Possible Publishers:

An authoritative, neutral body should manage the directory to ensure trust and adoption. Potential entities:

- **CMS:** Strong role in provider data and payer relationships; could integrate with NPPES or PECOS.
- **ONC:** Logical leader as ONC already maintains interoperability standards and coordinates TEFCA.
- **A Public-Private Partnership:** for example, the Sequoia Project – Offers flexibility and innovation while retaining federal oversight.
- **HINs/QHINs under TEFCA:** Qualified Health Information Networks could federate the directory under ONC oversight.

Should Users Bear a Cost? : We think the basic access ((e.g., endpoint lookup) should be open to promote equity and interoperability. Value added services could be priced such as Real-time API access, bulk data queries, or enriched metadata might incur a fee for vendors or enterprise users. These money collected could fund the maintenance of the directory.

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

The U.S. Core Data for Interoperability (USCDI) has contributed meaningful advancements in standardizing data exchange across healthcare systems; however, significant limitations persist in both its definitional scope and practical implementation. While USCDI establishes a foundational structure for interoperable health information, its current set of data elements falls short of fully supporting critical use cases such as social determinants of health, mental health workflows, and

care coordination. These gaps, combined with inconsistent vendor adoption and interpretation, hinder true interoperability and lead to variability in data quality, completeness, and exchange performance. To address these limitations, CORMAC recommends a hybrid approach that preserves structured USCDI elements while introducing standardized semi-structured formats and AI-friendly enhancements. This strategy balances the need for reliable, computable data with the flexibility required to accommodate evolving data types, ultimately improving data accessibility and clinical utility for all stakeholders.

a. Does it contain the full extent of data elements you need?

No, substantial limitations remain in USCDI coverage. Notable gaps include social determinants of health with only approximately 30% capture rate, mental health workflows that create implementation barriers, and care coordination data where automated exchanges often fail.

b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

The limitations stem from both definitional constraints in USCDI format and inconsistent utilization across vendors. Implementation variability presents significant challenges, with different EHR vendors interpreting USCDI requirements with 25-30% variance in code mappings, inconsistent handling of optional versus required elements, and dramatic performance variations ranging from 2 seconds to 45+ seconds for identical data requests.

c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

Adding more data elements could provide value but would create scoping challenges including increased implementation complexity, vendor compliance burden, and potential performance degradation. To address these challenges, a hybrid approach would be most effective: maintaining current USCDI structured elements as the foundation while adding standardized narrative fields with AI-extractable metadata tags and implementing machine-readable confidence scores for AI-extracted data elements. An incremental implementation strategy featuring tiered requirements with Core (mandatory), Enhanced (recommended), and Specialty (optional) categories would help manage complexity.

d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

Regarding less structured formats with language model processing, this approach could improve data coverage but would introduce new challenges including processing variability, quality assurance complexity, and potential inconsistency in data interpretation. A hybrid model combining structured USCDI elements with AI-processable narrative components would likely provide optimal balance between coverage and reliability while maintaining interoperability standards. We have designed properly, semi-structured formats (JSON with coded segments and free-text fields) allow NLP or large language models (LLMs) to extract meaning even when structured inputs are lacking. This approach may be particularly useful in surfacing data from PDFs, scanned forms, or clinical notes, which are often excluded from standard API responses but carry high patient value.

TD-9. Regarding certification of health IT:

c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

The government should introduce API coverage traceability requirements, where certified health IT systems must demonstrate that their APIs expose all available clinical content, including:

- Unstructured documents (via FHIR DocumentReference and Binary resources)
- Clinical notes (ie:JSON-formatted narrative text)
- Scanned documents and structured imaging metadata

Certification test scripts should validate field-level completeness across resource classes, and systems should be required to publish supported FHIR profiles, search capabilities, and known limitations. This would ensure that even if EHRs store diverse formats, those formats are not trapped from authorized external access.

d. What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data?

CMS could introduce API response completeness benchmarks as part of VBC and interoperability performance metrics, rewarding providers who maintain high coverage across clinical domains including labs, encounters, vitals, and care plans via FHIR-based responses. Similar to quality reporting incentives, this could be tied to:

- FHIR-based attestation fields submitted quarterly
- TEFCA participation thresholds
- Patient experience surveys measuring app access success rates

CMS could also align incentive programs to require participation in endpoint registries, allowing seamless discovery of active APIs across provider organizations.

4. Data Exchange

TD-12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

CMS should consider endorsing non-CMS data sources and networks—with proper guardrails. This will improve interoperability, expand patient access, and foster innovation in line with the 21st Century Cures Act and CMS's Digital Health Strategy.

Recommended Criteria & Metrics for CMS Endorsement

Category	Criteria or Metrics for CMS endorsement
Interoperability Standards|	Use of FHIR (US Core, Da Vinci), HL7, TEFCA participation
Data Accuracy & Completeness	Proven data quality, provenance tracking, timely updates
Privacy & Security	HIPAA compliance, robust consent frameworks, third-party security audits , and Section 508
Patient Access & Control	APIs that allow patients to access, download, and transmit their data easily
Performance Metrics	System uptime, query response time, scalability benchmarks
Transparency & Governance	Open governance model, transparent data-sharing policies, vendor neutrality
Public Benefit	demonstrable value to patients or providers

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

APIs that provide access to the entirety of a patient's Electronic Health Information (EHI)—as required under the 21st Century Cures Act—can unlock powerful opportunities for innovation, care improvement, and patient empowerment across the healthcare ecosystem.

a. What are the primary obstacles to this?

The following are some obstacles:

- Data & Standards: non-standard, unstructured EHI
- Information blocking, high fees, vendor lock-in
- Privacy & Trust: Data misuse by third-party apps
- Technical Limitations: Poor API reliability, incomplete historical data
- Patient Access: Digital divide, low tech literacy

b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?

The tradeoffs between using USCDI and full EHI stem from the balance between standardization and flexibility, as well as the technical, regulatory, and operational readiness of the healthcare ecosystem.

USCDI is optimized for real-time, standards-based interoperability and low-friction data exchange (e.g., patient access apps, EHR-to-EHR queries).. Full EHI opens the door to deeper, AI-driven use cases like comprehensive risk prediction, litigation support, or advanced research—but with greater complexity, liability, and compute costs. With today's advanced NLP, cloud computing, and data lakes, processing full EHI is more feasible than before, but still costly and uneven across healthcare systems.

5. Compliance

TD-18. Information blocking:

Could you, as a technology vendor, provide examples for the types of practices you have experienced that may constitute information blocking. Please include both situations of non-responsiveness as well as situations that may cause a failure or unusable response?

We've observed and experienced several **real-world practices that may constitute information blocking** under the 21st Century Cures Act. The following table summarizes them by category, some examples and the impact it creates.

Category	Example	Impact
Non-Responsiveness	Delayed FHIR API enablement	Prevents timely data access
Non-Responsiveness	Unaddressed integration requests	Stalls interoperability
Usability Failures	API throttling or uptime issues	Causes app failure
Usability Failures	Incomplete or filtered data	Undermines patient care
Usability Failures	Data sent in unusable formats	Obstructs automation or review
Procedural Barriers	Overuse of exceptions	Effectively blocks access